Second Order Derivatives

Exam code: 9MA0

Download PDF

Written by: Roger B
Updated on 20 June 2025

Did this video help you? Yes No

Help me understand

Second order derivatives

What is the second order derivative of a function?

- If you differentiate the derivative of a function (ie differentiate the function a second time) you get the **second order derivative** of the function
- For a function y = f(x), there are two forms of notation for the second derivative (or second order derivative)

$$f''(x) \text{ or } \frac{d^2y}{dx^2}$$

- Note the positions of the power of 2's in the second version

e.g. $f(x) = x^3$

$f'(x) = 3x^2$

$f''(x) = 6x$

$y = 7x^2 - 4x + 19$

$\frac{dy}{dx} = 14x - 4$

$\frac{d^2y}{dx^2} = 14$

Copyright © Save My Exams. All Rights Reserved

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

to as either the **first order derivative** or the **first derivative**

- The second order derivative gives the rate of change of the gradient function (ie of the first derivative) – this will be important for identifying the nature of **stationary points**

Examiner Tips and Tricks

- When finding second derivatives be especially careful with functions that have negative or fractional powers of ***x*** (see Worked Example below).
- Mistakes made with fractions or negative signs can build up as you calculate the derivative more than once.

Worked Example

Given that $f(x) = 4 - \sqrt{x} + \dfrac{3}{\sqrt{x}}$

a) Find $f'(x)$ and $f''(x)$.

b) Evaluate $f''(3)$. Give your answer in the form $a\sqrt{b}$, where b is an integer and a is a rational number.

a) $f(x) = 4 - x^{\frac{1}{2}} + 3x^{-\frac{1}{2}}$

REWRITE AS POWERS OF x

$f'(x) = -(\frac{1}{2})x^{\frac{1}{2}-1} + 3(-\frac{1}{2})x^{-\frac{1}{2}-1}$

$f'(x) = -\frac{1}{2}x^{-\frac{1}{2}} - \frac{3}{2}x^{-\frac{3}{2}}$

DIFFERENTIATE ONCE TO FIND $f'(x)$

$f''(x) = -\frac{1}{2}(-\frac{1}{2})x^{-\frac{1}{2}-1} - \frac{3}{2}(-\frac{3}{2})x^{-\frac{3}{2}-1}$

$f''(x) = \frac{1}{4}x^{\frac{3}{2}} + \frac{9}{4}x^{-\frac{5}{2}}$

DIFFERENTIATE A SECOND TIME TO FIND $f''(x)$

b) $f''(x) = \dfrac{1}{4x\sqrt{x}} + \dfrac{9}{4x^2\sqrt{x}}$

$x^{\frac{3}{2}} = x\sqrt{x}$ $\quad x^{\frac{5}{2}} = x^2\sqrt{x}$

$f''(3) = \dfrac{1}{12\sqrt{3}} + \dfrac{9}{36\sqrt{3}}$

$= \dfrac{12}{36\sqrt{3}} = \dfrac{1}{3\sqrt{3}} \times \dfrac{\sqrt{3}}{\sqrt{3}} = \dfrac{\sqrt{3}}{9}$

$f''(3) = \frac{1}{9}\sqrt{3}$

RATIONALISE DENOMINATOR

Copyright © Save My Exams. All Rights Reserved
save my exams

Teacher only

New! Exclusive Teacher-Only Questions

Fair accurate tests - built fast

Explore in Test Builder

Test yourself

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources

Learning Hub
Scholarship
Join
Past Papers
Solution Banks
Sitemap

Dashboard

Overview
Brand ambassador
Profile
Subscription
Settings
Log out

Company

About us
Exam Specificity
Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support

Subjects

Biology
Chemistry
Physics
Maths
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.